Wiki

This wiki serves as the central resource for setting up, managing, and optimizing your UNIT3D installation. Whether you're working on local development, managing a production server, or contributing to the open-source community, you'll find a number of useful guides here.

Backups

UNIT3D offers built in backup tools, available through the web dashboard or via Artisan commands, allowing you to create, manage, and restore your application snapshots.

1. Configuration

Customize `config/backup.php` in your editor and adjust settings as needed; inline notes outline the available configuration parameters.

Key structure:

- `backup`
  - `name`
  - `source`
    - `files`

      Specifies which directories and files to `include` and which to `exclude` in the backup.

```
'include' => [
    base_path(),
 ],

'exclude' => [
    base_path(),
    base_path('vendor'),
    base_path('node_modules'),
    base_path('storage'),
    base_path('public/vendor/joypixels'),
 ],
```

- `follow_links`
- `ignore_unreadable_directories`
- `relative_path`

- `databases`

  Specifies the database connections to back up.

- `database_dump_compressor`

  Compressor class (e.g. `Spatie\DbDumper\Compressors\GzipCompressor::class`) or `null` to disable.

- `destination`

  Defines the storage location for backup files.

- `temporary_directory`

Staging directory for temporary files.

- `notifications`

  Define when and how backup events trigger alerts via mail, Slack, or custom channels.

- `monitor_backups`

  Detect backup issues; triggers `UnhealthyBackupWasFound` when needed.

- `cleanup`

  Define how long to keep backups and when to purge old archives.

  - `strategy`
  - `default_strategy`

    Keeps all backups for 7 days; then retains daily backups for 16 days, weekly for 8 weeks, monthly for 4 months, and yearly for 2 years. Deletes old backups exceeding 5000 MB.

```
'keep_all_backups_for_days'                => 7,
'keep_daily_backups_for_days'              => 16,
'keep_weekly_backups_for_weeks'            => 8,
'keep_monthly_backups_for_months'          => 4,
'keep_yearly_backups_for_years'            => 2,
'delete_oldest_backups_when_using_more_megabytes_than' => 5000,
```

- `security`

  Ensure that only someone with your `APP_KEY` can decrypt and restore snapshots.

2. Create a backup

You can access the built-in Backups dashboard from the Staff menu. It shows each backup's status, health, size, and count, and lets administrators launch unscheduled full, database, or files-only backups instantly. Another approach is to use the command line.

> [!IMPORTANT] Backups initiated via the Staff Dashboard buttons may timeout on very large installations.

- The following artisan commands are available:

  ```
  php artisan backup:run
  ```

  Creates a timestamped ZIP of application files and database.

  ```
  php artisan backup:run --only-db
  ```

  Creates a timestamped ZIP containing only the database.

  ```
  php artisan backup:run --only-files
  ```

  Creates a timestamped ZIP containing only application files.

3. Viewing backup list

- **List** existing backups:

```
php artisan backup:list
```

4. Restoring a backup

[!WARNING] **Always test backup restoration procedures on a non-critical environment before applying to production.** Incorrect restoration can lead to data loss or service disruption.

1. **Install prerequisites** (Debian/Ubuntu):

```
sudo apt update && sudo apt install p7zip-full -y
```

2. **Retrieve** your application key:

```
grep '^APP_KEY=' /var/www/html/.env
```

3. **Extract** the outer archive (enter APP_KEY when prompted):

```
7z x [UNIT3D]YYYY-MM-DD-HH-MM-SS.zip
```

4. **Unzip** inner archive, if generated:

```
7z x backup.zip
```

[!NOTE] **Full backups will contain two parts; the files backup and a database backup or dump file.**

Restoring the files backup:

1. **Copy** restored files to webroot:

```
sudo cp -a ~/tempBackup/var/www/html/. /var/www/html/
```

2. **Fix** file permissions:

```
sudo chown -R www-data:www-data /var/www/html
sudo find /var/www/html -type f -exec chmod 664 {} \;
sudo find /var/www/html -type d -exec chmod 775 {} \;
```

Restoring the database backup:

1. **Retrieve** your database credentials:

```
grep '^DB_' /var/www/html/.env
```

2. **Restore** your database:

```
mysql -u unit3d -p unit3d < ~/tempBackup/db-dumps/mysql-unit3d.sql
```

5. Reset & Cleanup

After restoring backups, ensure to reset configurations and clean up temporary files to maintain system integrity.

1. **Fix** permissions:

```
sudo chown -R www-data:www-data /var/www/html
sudo find /var/www/html -type f -exec chmod 664 {} \;
sudo find /var/www/html -type d -exec chmod 775 {} \;
```

2. **Reinstall** Node dependencies & rebuild (if assets need it):

```
sudo rm -rf node_modules && sudo bun install && sudo bun run build
```

3. **Clear** caches & restart services:

```
cd /var/www/html
sudo php artisan set:all_cache
sudo systemctl restart php8.4-fpm
sudo php artisan queue:restart
```

Basic tuning

> [!IMPORTANT] These guides are intended for UNIT3D v8.0.0 + instances. While these are better than defaults be careful blindly following them. Proper tuning requires understanding your server, running tests and monitoring the results.

Redis single server

Category	Severity	Time To Fix
:rocket: Performance	Major	30 minutes

Introduction

If your Redis service is running on your web server, it is highly recommended that you use **Unix sockets** instead of **TCP ports** to communicate with your web server.

Based on the Redis official benchmark, you can **improve performance by up to 50%** using unix sockets (versus TCP ports) on Redis.

Of course, unix sockets can only be used if both your Laravel application and Redis are running on the same server.

How to enable unix sockets

First, create the redis folder that the unix socket will be in and set appropriate permissions:

```
sudo mkdir -p /var/run/redis/
sudo chown -R redis:www-data /var/run/redis
sudo usermod -aG redis www-data
```

Next, add the unix socket path and permissions in your Redis configuration file (typically at `/etc/redis/redis.conf`):

```
unixsocket /var/run/redis/redis.sock
unixsocketperm 770
```

Finally, set your corresponding env variables to the socket path as above:

```
REDIS_HOST=/var/run/redis/redis.sock
REDIS_PORT=-1
REDIS_SCHEME=unix
```

Ensure that you have your `config/database.php` file refer to the above variables (notice the `scheme` addition below):

```
'redis' => [
    'client' => env('REDIS_CLIENT', 'phpredis'),

    'options' => [
        'scheme' => env('REDIS_SCHEME', 'tcp'),
    ],

    'default' => [
        'url' => env('REDIS_URL'),
        'host' => env('REDIS_HOST', '127.0.0.1'),
        'password' => env('REDIS_PASSWORD', null),
        'port' => env('REDIS_PORT', '6379'),
        'database' => env('REDIS_DB', '0'),
    ],

    'cache' => [
        'url' => env('REDIS_URL'),
        'host' => env('REDIS_HOST', '127.0.0.1'),
        'password' => env('REDIS_PASSWORD', null),
        'port' => env('REDIS_PORT', '6379'),
        'database' => env('REDIS_CACHE_DB', '1'),
    ],
],
```

Once that's all done simply restart redis.

```
sudo systemctl restart redis
```

References

- Redis Official Benchmark on Socket vs TCP
- Laravel Documentation on Redis

[!NOTE] Keep in mind that when using unix socket you will now connect to redis-cli in terminal like so: `redis-cli -s /var/run/redis/redis.sock`

MySQL single server

Category	Severity	Time To Fix
:rocket: Performance	Major	10 minutes

Introduction

If your MySQL database is running on your web server, it is highly recommended that you use **Unix sockets** instead of **TCP ports** to communicate with your web server.

Based on Percona's benchmark, you can **improve performance by up to 50%** using unix sockets (versus TCP ports on MySQL).

Of course, unix sockets can only be used if both your UNIT3D application and database are running on the same server which is by default.

How to enable unix sockets

First, open your MySQL configuration file.

```
nano /etc/mysql/my.cnf
```

Then, uncomment and change (if needed) the socket parameter in the `[mysqld]` section of one of the above configuration files:

```
[mysqld]
user            = mysql
pid-file        = /var/run/mysqld/mysqld.pid
socket          = /var/run/mysqld/mysqld.sock
port            = 3306
```

Close this file, then ensure that the mysqld.sock file exists by running an ls command on the directory where MySQL expects to find it:

```
ls -a /var/run/mysqld/
```

If the socket file exists, you will see it in this command's output:

Output `. .. mysqld.pid mysqld.sock mysqld.sock.lock` If the file does not exist, the reason may be that MySQL is trying to create it, but does not have adequate permissions to do so. You can ensure that the correct permissions are in place by changing the directory's ownership to the mysql user and group:

```
sudo chown mysql:mysql /var/run/mysqld/
```

Then ensure that the mysql user has the appropriate permissions over the directory. Setting these to 775 will work in most cases:

```
sudo chmod -R 755 /var/run/mysqld/
```

Finally, set your `database.connections.mysql.unix_socket` configuration variable or the corresponding env variable to the socket path as above:

```
DB_SOCKET=/var/run/mysqld/mysqld.sock
```

Once that's all done simply refresh your cache and then restart the services.

```
php artisan set:all_cache
```

```
sudo systemctl restart mysql && sudo systemctl restart php8.3-fpm && sudo systemctl
restart nginx
```

References

- Percona Benchmark (Unix vs TCP)
- MySQL Unix Socket Setup
- MySQL Shell Connections Guide

Composer autoloader optimization

Category	Severity	Time To Fix
:rocket: Performance	Moderate	5 minutes

Introduction

Due to the way PSR-0/4 autoloading rules are defined, the Composer autoloader checks the filesystem before resolving a classname conclusively.

In production, Composer allows for optimization to convert the PSR-0 and PSR-4 autoloading rules into classmap rules, making autoloading quite a bit faster. In production we also don't need all the

require-dev dependencies loaded up!

How to optimize?

It's really simple. SSH to your server and run the following commands.

```
composer install --prefer-dist --no-dev
```

```
composer dump-autoload --optimize
```

References

- https://getcomposer.org/doc/articles/autoloader-optimization.md

PHP8 OPCache

Category	Severity	Time To Fix
:rocket: Performance	Major	10 minutes

Introduction

Opcache provides massive performance gains. One of the benchmarks suggest it can provide a 5.5X performance gain in a Laravel application!

`What is OPcache?` Every time you execute a PHP script, the script needs to be compiled to byte code. OPCache leverages a cache for this bytecode, so the next time the same script is requested, it doesn't have to recompile it. This can save some precious execution time, and thus make UNIT3D faster.

`Sounds awesome, so how can you use it?` Easy. SSH to your server and run the following command. `sudo nano /etc/php/8.3/fpm/php.ini` This is assuming your on PHP 8.3. If not then adjust the command. Once you have the config open search for `opcache`.

Now you can change some values, I will walk you through the most important ones.

`opcache.enable=1` This of course, enables OPcache for php-fpm. Make sure it is uncommented. AKA remove the `;`

`opcache.enable_cli=1` This of course, enables OPcache for php-cli. Make sure it is uncommented. AKA remove the `;`

`opcache.memory_consumption=256M` How many Megabyte you want to assign to OPCache. Choose anything higher than 64 (default value) depending on your needs. 2GB is sufficient but if have more RAM then make use of it! Make sure it is uncommented. AKA remove the `;`

`opcache.interned_strings_buffer=64` How many Megabyte you want to assign to interned strings. Choose anything higher than 16 (default value). 1GB is sufficient but if have more RAM then make use of it! Make sure it is uncommented. AKA remove the `;`

`opcache.validate_timestamps=0` This will revalidate the script. If you set this to 0 (best performance), you need to manually clear the OPcache every time your PHP code changes. So if you update UNIT3D using `php artisan git:update` or manually make changes yourself you need to run `sudo systemctl restart php8.2-fpm` afterwords for your changes to take effect and show. Make sure it is uncommented. AKA remove the `;`

`opcache.save_comments=1` This will preserve comments in your script, I recommend to keep this enabled, as some libraries depend on it, and I couldn't find any benefits from disabling it (except from saving a few bytes RAM). Make sure it is uncommented. AKA remove the `;`

And there you have it folks. Experiment with these values, depending on the resources of your server. Save the file and exit and restart PHP `sudo systemctl restart php8.3-fpm`.

Enjoy! 🖐

PHP 8 preloading

Category	Severity	Time To Fix
:rocket: Performance	Major	5 minutes

Introduction

This is chaining off `Want More Performance? Lets talk about OPCache!` guide. You must have OPCache enabled to use preloading.

PHP preloading for PHP >=7.4. Preloading is a feature of php that will pre-compile php functions and classes to opcache. Thus, this becomes available in your programs with out needing to require the files, which improves speed. To read more on php preloading you can see the opcache.preloading documentation.

Enabling preloading

SSH to your server and run the following command. `sudo nano /etc/php/8.3/fpm/php.ini` This is assuming your on PHP 8.3. If not then adjust the command. Once you have the config open search for `preload` .

Now you can change some values.

```
; Specifies a PHP script that is going to be compiled and executed at server
; start-up.
; https://php.net/opcache.preload
opcache.preload = '/var/www/html/preload.php';

; Preloading code as root is not allowed for security reasons. This directive
; facilitates to let the preloading to be run as another user.
; https://php.net/opcache.preload_user
opcache.preload_user=ubuntu
```

As you can see we are calling the preload file included in UNIT3D located in `/var/www/html/preload.php` . `opcache.preload_user=ubuntu` you should changed to your server user. Not root!!!!

And there you have it folks. Save the file and exit and restart PHP `sudo systemctl restart php8.3-fpm` . You are now preloading Laravel thus making UNIT3D faster.

PHP8 JIT

Category	Severity	Time To Fix
:rocket: Performance	Moderate	5 minutes

Introduction

PHP 8 adds a JIT compiler to PHP's core which has the potential to speed up performance dramatically.

First of all, the JIT will only work if opcache is enabled, this is the default for most PHP installations, but you should make sure that `opcache.enable` is set to `1` in your php.ini file. Enabling the JIT itself is done by specifying `opcache.jit_buffer_size` in php.ini. ***So I recommend checking the OPcache guide I made first then coming back here.***

How to enable JIT

SSH to your server and run the following command. `sudo nano /etc/php/8.3/fpm/php.ini` This is assuming your on PHP 8.2. If not then adjust the command. Once you have the config open search for `opcache.jit`.

If you do not get any results then search for `[curl]` you should see the following.

```
[curl]
; A default value for the CURLOPT_CAINFO option. This is required to be an
; absolute path.
;curl.cainfo =
```

Right above it add:

```
opcache.jit_buffer_size=256M
```

Its as simple as that. Save and exit and restart PHP. `sudo systemctl restart php8.2-fpm`

PM static

Category	Severity	Time To Fix
:rocket: Performance	Major	10 minutes

> [!IMPORTANT] This guide is intended for high traffic sites.

Introduction

Lets give a basic description on what these options are:

`pm = dynamic` – the number of child processes is set dynamically based on the following directives: pm.max_children, pm.start_servers,pm.min_spare_servers, pm.max_spare_servers.

`pm = ondemand` – the processes spawn on demand (when requested, as opposed to dynamic, where pm.start_servers are started when the service is started.

`pm = static` – the number of child processes is fixed by pm.max_children.

The PHP-FPM pm static setting depends heavily on how much free memory your server has. Basically if you are suffering from low server memory, then pm ondemand or dynamic maybe be better options. On the other hand, if you have the memory available you can avoid much of the PHP process manager (PM) overhead by setting pm static to the max capacity of your server. In other words, when you do the math, pm.static should be set to the max amount of PHP-FPM processes that can run without creating memory availability or cache pressure issues. Also, not so high as to overwhelm CPU(s) and have a pile of pending PHP-FPM operations.

Enabling static

Lets open up our PHP configuration file. `sudo nano /etc/php/8.3/fpm/pool.d/www.conf`

Set `pm = static` Set `pm.max_children = 25`

Save, Exit and Restart `sudo systemctl restart php8.3-fpm`

Conclusion

When it comes to PHP-FPM, once you start to serve serious traffic, ondemand and dynamic process managers for PHP-FPM can limit throughput because of the inherent overhead. Know your system and set your PHP-FPM processes to match your server's max capacity.

UNIT3D development on Arch Linux with Laravel Sail

A guide by EkoNesLeg

This guide outlines the steps to set up UNIT3D using Laravel Sail on Arch Linux. While the guide highlights the use of Arch Linux, the instructions can be adapted to other environments.

> [!IMPORTANT] This guide is intended for local development environments only and is not suitable for production deployment.

Modifying `.env` and secure headers for non-HTTPS instances

For local development, HTTP is commonly used instead of HTTPS. To prevent mixed content issues, adjust your `.env` file as follows:

1. **Modify the `.env` Config:**
    - Open your `.env` file in the root directory of your UNIT3D project.
    - Add or modify the following environment variables:

```
DB_HOST=mysql               # Match the container name in the compose file
DB_USERNAME=unit3d          # The username can be anything except `root`
SESSION_SECURE_COOKIE=false # Disables secure cookies
REDIS_HOST=redis            # Match the container name in the compose file
CSP_ENABLED=false           # Disables Content Security Policy
HSTS_ENABLED=false          # Disables Strict Transport Security
```

Prerequisites

Ensure Docker and Docker Compose are installed, as they are required for managing the Dockerized development environment provided by Laravel Sail.

Installing Docker and Docker Compose

Refer to the Arch Linux Docker documentation and install the necessary packages:

```
sudo pacman -S docker docker-compose
```

Step 1: clone the repository

Clone the UNIT3D repository to your local environment:

1. Navigate to your chosen workspace directory:

```
cd ~/PhpstormProjects
```

2. Clone the repository:

```
git clone git@github.com:HDInnovations/UNIT3D-Community-Edition.git
```

Step 2: Composer dependency installation

1. **Change to the project's root directory:**

```
cd ~/PhpstormProjects/UNIT3D-Community-Edition
```

2. **Install Composer dependencies:**

   Run the following command to install the PHP dependencies:

```
composer install
```

Step 3: Docker environment initialization

1. **Switch to branch `development`:**

   Before starting Docker, switch to the `development` branch:

```
git checkout development
```

2. **Start the Docker environment using Laravel Sail:**

```
./vendor/bin/sail up -d
```

Step 4: app key generation

Generate a new `APP_KEY` in the `.env` file for encryption:

```
./vendor/bin/sail artisan key:generate
```

Note: If you are importing a database backup, make sure to set the `APP_KEY` in the `.env` file to match the key used when the backup was created.

Step 5: database migrations and seeders

Initialize your database with sample data by running migrations and seeders:

```
./vendor/bin/sail artisan migrate:fresh --seed
```

> [!IMPORTANT] This operation resets your database and seeds it with default data. Avoid running this in a production environment.

Step 6: database preparation

Initial database loading

Prepare your database with the initial schema and data. Make sure you have a database dump file, such as `prod-site-backup.sql`.

MySQL data importation

Import your database dump into MySQL within the Docker environment:

```
./vendor/bin/sail mysql -u root -p unit3d < prod-site-backup.sql
```

Note: Ensure that the `APP_KEY` in the `.env` file matches the key used in your deployment environment for compatibility.

Step 7: NPM dependency management

Manage Node.js dependencies and compile assets within the Docker environment:

```
./vendor/bin/sail bun install
./vendor/bin/sail bun run build
```

If needed, refresh the Node.js environment:

```
./vendor/bin/sail rm -rf node_modules && bun pm cache rm && bun install && bun run build
```

Step 8: application cache configuration

Optimize the application's performance by setting up the cache:

```
./vendor/bin/sail artisan set:all_cache
```

Step 9: environment restart

Apply new configurations or restart the environment by toggling the Docker environment:

```
./vendor/bin/sail restart && ./vendor/bin/sail artisan queue:restart
```

Additional notes

- **Permissions**: Use `sudo` cautiously to avoid permission conflicts, particularly with Docker commands that require elevated access.

Appendix: Sail commands for UNIT3D

This section provides a reference for managing and interacting with UNIT3D using Laravel Sail.

Docker management

- **Start environment**:

  ```
  ./vendor/bin/sail up -d
  ```

  Starts Docker containers in detached mode.

- **Stop environment**:

  ```
  ./vendor/bin/sail down
  ```

  Stops and removes Docker containers.

- **Restart environment**:

  ```
  ./vendor/bin/sail restart
  ```

  Applies changes by restarting the Docker environment.

Dependency management

- **Install Composer dependencies**:

  ```
  ./vendor/bin/sail composer install
  ```

  Installs PHP dependencies defined in `composer.json`.

- **Update Composer dependencies**:

  ```
  ./vendor/bin/sail composer update
  ```

  Updates PHP dependencies defined in `composer.json`.

Laravel Artisan

- **Run migrations**:

  ```
  ./vendor/bin/sail artisan migrate
  ```

  Executes database migrations.

- **Seed database**:

  ```
  ./vendor/bin/sail artisan db:seed
  ```

  Seeds the database with predefined data.

- **Refresh database**:

  ```
  ./vendor/bin/sail artisan migrate:fresh --seed
  ```

  Resets and seeds the database.

- **Cache configurations**:

  ```
  ./vendor/bin/sail artisan set:all_cache
  ```

Clears and caches configurations for performance.

NPM and assets

- **Install NPM dependencies**:

  ```
  ./vendor/bin/sail bun install
  ```

  Installs Node.js dependencies.

- **Compile assets**:

  ```
  ./vendor/bin/sail bun run build
  ```

  Compiles CSS and JavaScript assets.

Database operations

- **MySQL interaction**:

  ```
  ./vendor/bin/sail mysql -u root -p
  ```

  Opens MySQL CLI for database interaction.

Queue management

- **Restart queue workers**:

```
./vendor/bin/sail artisan queue:restart
```

Restarts queue workers after changes.

Troubleshooting

- **View logs**:

  ```
  ./vendor/bin/sail logs
  ```

  Displays Docker container logs.

- **Run PHPUnit (PEST) tests**:

  ```
  ./vendor/bin/sail artisan test
  ```

  Runs PEST tests for the application.

UNIT3D v8.x.x on MacOS with Laravel Sail and PhpStorm

A guide by HDVinnie

This guide is designed for setting up UNIT3D, a Laravel application, leveraging Laravel Sail on MacOS.

Warning: This setup guide is intended for local development environments only and is not suitable for production deployment.

Modifying .env and secure headers for non-HTTPS instances

For local development, it's common to use HTTP instead of HTTPS. To prevent mixed content issues, follow these steps:

1. **Modify the `.env` config:**
   - Open your `.env` file located in the root directory of your UNIT3D project.
   - Find the `SESSION_SECURE_COOKIE` setting and change its value to `false`. This action disables secure cookies, which are otherwise required for HTTPS.

```
SESSION_SECURE_COOKIE=false
```

2. **Adjust the secure headers in `config/secure-headers.php`:**
   - Navigate to the `config` directory and open the `secure-headers.php` file.
   - To disable the `Strict-Transport-Security` header, locate the `hsts` setting and change

its value to `false`.

```
'hsts' => false,
```

- Next, locate the Content Security Policy (CSP) configuration to adjust it for HTTP. Disable the CSP to prevent it from blocking content that doesn't meet the HTTPS security requirements.

```
'enable' => env('CSP_ENABLED', false),
```

Prerequisites

Installing Docker Desktop

Install Docker Desktop

Once installed, launch Docker Desktop

Installing GitHub Desktop

Install GitHub Desktop

Once installed, launch GitHub Desktop

Installing PHPStorm

Install PHPStorm

Once installed, launch PHPStorm

Step 1: clone the repository

Firstly, clone the UNIT3D repository to your local environment by visiting UNIT3D-Community-Edition Repo. Then click the blue colored code button and select `Open with Github Desktop`. Once Github Desktop is open set you local path to clone to like `/Users/HDVinnie/Documents/Personal/UNIT3D-Community-Edition`

Step 2: open the project in PHPStorm

Within PHPStorm goto `File` and then click `Open`. Select the local path you just did like `/Users/HDVinnie/Documents/Personal/UNIT3D-Community-Edition`.

The following commands are run in PHPStorm. Can do so by clicking `Tools->Run Command`.

Step 3: start Sail

Initialize the Docker environment using Laravel Sail:

```
./vendor/bin/sail up -d
```

Step 4: Composer dependency installation

```
./vendor/bin/sail composer install
```

Step 5: Bun dependency install and compile assets

```
./vendor/bin/sail bun install
```

```
./vendor/bin/sail bun run build
```

Step 6: database migrations and seeders

For database initialization with sample data, apply migrations and seeders:

```
./vendor/bin/sail artisan migrate:fresh --seed
```

Caution: This operation will reset your database and seed it with default data. Exercise caution in production settings.

Step 7: database preparation (if want to use a production database backup locally)

Initial database loading

Prepare your database with the initial schema and data. Ensure you have a database dump file, e.g., `prod-site-backup.sql`.

MySQL data importation

To import your database dump into MySQL within the local environment, use:

```
./vendor/bin/sail mysql -u root -p unit3d < prod-site-backup.sql
```

Note: For this to work properly you must set the APP_KEY value in your local `.env` file to match you prod APP_KEY value.

Step 8: application cache configuration

Optimize the application's performance by setting up the cache:

```
sail artisan set:all_cache
```

Step 9: visit local instance

Open your browser and visit `localhost`. Enjoy!

Additional notes

- **Permissions**: Exercise caution with `sudo` to avoid permission conflicts, particularly for Docker commands requiring elevated access.

Appendix: Sail commands for UNIT3D

This section outlines commands for managing and interacting with UNIT3D using Laravel Sail.

Sail management

- **Start environment**:

  ```
  ./vendor/bin/sail up -d
  ```

  Starts Docker containers in detached mode.

- **Stop environment**:

  ```
  ./vendor/bin/sail down -v
  ```

  Stops and removes Docker containers.

- **Restart environment**:

  ```
  ./vendor/bin/sail restart
  ```

  Applies changes by restarting Docker environment.

Dependency management

- **Install Composer dependencies**:

  ```
  ./vendor/bin/sail composer install
  ```

  Installs PHP dependencies defined in `composer.json`.

- **Update Composer dependencies**:

  ```
  ./vendor/bin/sail composer update
  ```

  Updates PHP dependencies defined in `composer.json`.

Laravel Artisan

- **Run migrations**:

  ```
  ./vendor/bin/sail artisan migrate
  ```

  Executes database migrations.

- **Seed database**:

```
./vendor/bin/sail artisan db:seed
```

Seeds database with predefined data.

- **Refresh database**:

```
./vendor/bin/sail artisan migrate:fresh --seed
```

Resets and seeds database.

- **Cache configurations**:

```
./vendor/bin/sail artisan set:all_cache
```

Clears and caches configurations for performance.

NPM and assets

- **Install Bun dependencies**:

```
./vendor/bin/sail bun install
```

Installs Node.js dependencies.

- **Compile assets**:

```
./vendor/bin/sail bun run build
```

Compiles CSS and JavaScript assets.

Database operations

- **MySQL interaction**:

  ```
  ./vendor/bin/sail mysql -u root -p
  ```

  Opens MySQL CLI for database interaction.

Queue management

- **Restart queue workers**:

  ```
  ./vendor/bin/sail artisan queue:restart
  ```

  Restarts queue workers after changes.

Troubleshooting

- **View logs**:

  ```
  ./vendor/bin/sail logs
  ```

  Displays Docker container logs.

- **PHPUnit (PEST) tests**:

  ```
  ./vendor/bin/sail artisan test
  ```

Runs PEST tests for application.

Meilisearch setup for UNIT3D

Note: This guide assumes you are using a `sudo` user named `ubuntu`.

1. Install and configure Meilisearch

1. **Install Meilisearch:**

```
sudo curl -L https://install.meilisearch.com | sudo sh
sudo mv ./meilisearch /usr/local/bin/
sudo chmod +x /usr/local/bin/meilisearch
```

2. **Set up directories:**

```
sudo mkdir -p /var/lib/meilisearch/data /var/lib/meilisearch/dumps /var/lib/
meilisearch/snapshots
sudo chown -R ubuntu:ubuntu /var/lib/meilisearch
sudo chmod -R 750 /var/lib/meilisearch
```

3. **Generate and record a master key:**

   Generate a 16-byte master key:

```
openssl rand -hex 16
```

   Record this key, as it will be used in the configuration file.

4. **Configure Meilisearch:**

```
sudo curl https://raw.githubusercontent.com/meilisearch/meilisearch/latest/
config.toml -o /etc/meilisearch.toml
sudo nano /etc/meilisearch.toml
```

Update the following in `/etc/meilisearch.toml`:

```
env = "production"
master_key = "your_16_byte_master_key"
db_path = "/var/lib/meilisearch/data"
dump_dir = "/var/lib/meilisearch/dumps"
snapshot_dir = "/var/lib/meilisearch/snapshots"
```

5. **Create and enable service:**

```
sudo nano /etc/systemd/system/meilisearch.service
```

Add the following:

```
[Unit]
Description=Meilisearch
After=systemd-user-sessions.service

[Service]
Type=simple
WorkingDirectory=/var/lib/meilisearch
ExecStart=/usr/local/bin/meilisearch --config-file-path /etc/meilisearch.toml
User=ubuntu
Group=ubuntu
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

Enable and start the service:

```
sudo systemctl enable meilisearch
sudo systemctl start meilisearch
sudo systemctl status meilisearch
```

2. Configure UNIT3D for Meilisearch

1. **Update `.env`:**

```
sudo nano /var/www/html/.env
```

Add the following:

```
SCOUT_DRIVER=meilisearch
MEILISEARCH_HOST=http://127.0.0.1:7700
MEILISEARCH_KEY=your_16_byte_master_key
```

2. **Clear configuration and restart services:**

```
sudo php artisan set:all_cache
sudo systemctl restart php8.3-fpm
sudo php artisan queue:restart
```

3. Maintenance

1. **Reload data and sync indexes:**

   - **Sync index settings:**

     After UNIT3D updates, sync the index settings to ensure they are up to date:

     ```
     sudo php artisan scout:sync-index-settings
     ```

   - **Reload data:**

     Whenever Meilisearch is upgraded or during the initial setup, the database must be reloaded:

```
sudo php artisan auto:sync_torrents_to_meilisearch --wipe && sudo php artisan
auto:sync_people_to_meilisearch
```

See also

For further details, refer to the official Meilisearch documentation.

Server management

[!IMPORTANT] The following assumptions are made:

- You have one `root` user and one regular user with sudo privileges on the dedicated server.
- The regular user with sudo privileges is assumed to have the username `ubuntu`.
- The project root directory is located at `/var/www/html`.
- All commands are run from the project root directory.

1. Elevated shell

All SSH and SFTP operations should be conducted using the non-root user. Use `sudo` for any commands that require elevated privileges. Do not use the `root` user directly.

2. File permissions

Ensure that everything in `/var/www/html` is owned by `www-data:www-data`, except for `node_modules`, which should be owned by `root:root`.

Set up these permissions with the following commands:

```
sudo usermod -a -G www-data ubuntu
sudo chown -R www-data:www-data /var/www/html
sudo find /var/www/html -type f -exec chmod 664 {} \;
sudo find /var/www/html -type d -exec chmod 775 {} \;
sudo chgrp -R www-data storage bootstrap/cache
sudo chmod -R ug+rwx storage bootstrap/cache
sudo rm -rf node_modules && sudo bun install && sudo bun run build
```

3. Handling code changes

PHP changes

If any PHP files are modified, run the following commands to clear the cache, restart the PHP-FPM service, and restart the Laravel queues:

```
sudo php artisan set:all_cache && sudo systemctl restart php8.3-fpm && sudo php
artisan queue:restart
```

Static assets (SCSS, JS)

If you make changes to SCSS or JavaScript files, rebuild the static assets using:

```
bun run build
```

4. Changing the domain

1. **Update the environment variables:**

   Modify the domain in the `APP_URL` and `MIX_ECHO_ADDRESS` variables within the `.env` file:

   ```
   sudo nano ./.env
   ```

2. **Refresh the TLS certificate:**

   Use `certbot` to refresh the TLS certificate:

   ```
   certbot --redirect --nginx -n --agree-tos --email=sysop@your_domain.tld -d
   your_domain.tld -d www.your_domain.tld --rsa-key-size 2048
   ```

3. **Update the WebSocket configuration:**

   Update all domains listed in the WebSocket configuration to reflect the new domain:

   ```
   sudo nano ./laravel-echo-server.json
   ```

4. **Restart the chatbox server:**

   Reload the Supervisor configuration to apply changes:

   ```
   sudo supervisorctl reload
   ```

5. **Compile static assets:**

   Rebuild the static assets:

```
bun run build
```

5. Meilisearch maintenance

Refer Meilisearch setup for UNIT3D, specifically the maintenance section, for managing upgrades and syncing indexes.

UNIT3D open source: how to share your source code

A guide by EkoNesLeg

1. Introduction

As part of complying with the GNU Affero General Public License (AGPL), sites that modify and distribute UNIT3D are required to share their source code. This guide provides an easy process for creating a sanitized tarball of your modified source code and encourages you to create and update an "Open Source" page on your site to make this code available.

2. Setting up tarball creation

2.1 Exclude sensitive files

To create a tarball that includes only the modified source code and excludes sensitive files like configuration data, you can take advantage of the existing `.gitignore` file in your UNIT3D deployment. Here's how:

1. **Reference `.gitignore` for exclusions:**

   If your production environment has the original `.gitignore` file that already lists the files and

directories you don’t want to include in version control, you can use it to exclude those same items from your tarball:

```
ln -s /var/www/html/.gitignore /var/www/html/.tarball_exclude
```

2. **Additional exclusions (if needed):**

   If additional exclusions are needed, or if you've removed the git environment from your production environment, you should manually add the exclusions to the `.tarball_exclude` file:

   ```
   nano /var/www/html/.tarball_exclude
   ```

   Add the following to the file:

```
.env
node_modules
storage
vendor
public
*.gz
*.lock
UNIT3D-Announce
unit3d-announce
laravel-echo-server.json
config
.DS_Store
.idea
.vscode
nbproject
.phpunit.cache
.ftpconfig
storage/backups
storage/debugbar
storage/gitupdate
storage/*.key
laravel-echo-server.lock
.vagrant
Homestead.json
Homestead.yaml
npm-debug.log
_ide_helper.php
supervisor.ini
.phpunit.cache/
.phpstan.cache/
caddy
frankenphp
frankenphp-worker.php
data
config/caddy/autosave.json
build
bootstrap
*.sql
```

```
*.DS_Storecomposer.lock
*.swp
coverage.xml
cghooks.lock
*.pyc
emojipy-*
emojipy.egg-info/
lib/js/tests.html
lib/js/tests/npm-debug.log
```

2.2 Create the tarball

1. **Create a script to generate the tarball:**

```
nano /var/www/html/create_tarball.sh
```

2. **Add the following content to the script:**

```
#!/bin/bash

TARBALL_NAME="UNIT3D_Source_$(date +%Y%m%d_%H%M%S).tar.gz"
TAR_EXCLUDES="--exclude-from=/var/www/html/.tarball_exclude"

tar $TAR_EXCLUDES -czf /var/www/html/public/$TARBALL_NAME -C /var/www html

# Create a symlink to the latest tarball
ln -sf "/var/www/html/public/$TARBALL_NAME" "/var/www/html/public/
UNIT3D_Source_LATEST.tar.gz"
```

3. **Make the script executable:**

```
chmod +x /var/www/html/create_tarball.sh
```

4. **Run the script manually whenever you update your site:**

```
/var/www/html/create_tarball.sh
```

3. Creating and updating the "Open source" Page

1. **Create an "Open source" page:**

   Go to your site's `/dashboard/pages` section and create a new page called "Open source."

2. **Add the following Markdown content to the page:**

```
## Open source

We comply with the UNIT3D's GNU Affero General Public License (AGPL) by sharing
our modified source code. You can download the latest version of our source code
below.

- **[Download Latest Source Code](/UNIT3D_Source_LATEST.tar.gz)**

### License information

Our site runs on a modified version of [UNIT3D](https://github.com/HDInnovations/
UNIT3D-Community-Edition). For more details on the license, visit the [GNU AGPL
License](https://github.com/HDInnovations/UNIT3D-Community-Edition/blob/master/
LICENSE).
```

3. **Manually update this page whenever you update your site:**

   After running the tarball creation script, update the page content if necessary to reflect any changes or additional notes about the modifications made.

4. Encouraging compliance and contributions

By publicly sharing your modified source code, you not only comply with the AGPL but also contribute to the open-source community. We encourage sites to contribute their changes back to the upstream repository by submitting pull requests, which helps improve UNIT3D for everyone.

Torrent API

Introduction

UNIT3D is offering a new `{JSON:API}`. If you haven't already head over to your profile. Hover Settings nav and click API Key. From there you can generate or reset your API Key.

Ecosystem

Torrent auto uploaders

- L4G's Upload Assistant — a simple tool to take the work out of uploading
- GG Bot Upload Assistant — a torrent auto uploader to take the manual work out of uploading

API authentication

There are several ways of passing the API token to UNIT3D. We'll discuss each of these approaches while using the Guzzle HTTP library to demonstrate their usage. You may choose any of these approaches based on your needs to communicate with our API.

- **Query string**

UNIT3D's API consumers may specify their token as an `api_token` query string value:

```
$response = $client->request('GET', '/api/torrents?api_token=YOUR_TOKEN_HERE);
```

- **Request payload**

  UNIT3D's API consumers may include their API token in the request's form parameters as an `api_token`:

```
$response = $client->request('POST', '/api/torrents', [
    'headers' => [
        'Accept' => 'application/json',
    ],
    'form_params' => [
        'api_token' => 'YOUR_TOKEN_HERE',
    ],
]);
```

- **Bearer token**

  UNIT3D's API consumers may provide their API token as a `Bearer` token in the `Authorization` header of the request:

```
$response = $client->request('POST', '/api/torrents', [
    'headers' => [
        'Authorization' => 'Bearer YOUR_TOKEN_HERE',
        'Accept' => 'application/json',
    ],
]);
```

API endpoints

Upload a torrent

Endpoint: POST `/api/torrents/upload`

Parameters:

Parameter	Type	Description
`torrent`	file	.torrent file
`nfo`	file	.nfo file
`name`	string	Torrent name
`description`	string	Torrent description
`mediainfo`	string	MediaInfo text output
`bdinfo`	string	BDInfo quick summary output
`category_id`	int	Category ID
`type_id`	int	Type ID
`resolution_id`	int	Resolution ID
`region_id`	int	Region ID
`distributor_id`	int	Distributor ID
`season_number`	int	Season number (TV only)
`episode_number`	int	Episode number (TV only)
`tmdb`	int	TMDB ID
`imdb`	int	IMDB ID

Parameter	Type	Description
`tvdb`	int	TVDB ID
`mal`	int	MAL ID
`igdb`	int	IGDB ID (Games only)
`anonymous`	bool	Should the uploader's username be hidden?
`personal_release`	bool	Is the torrent's content created by the uploader?
`internal` *	bool	Is the torrent an internal release?
`refundable` *	bool	Is the torrent refundable?
`featured` *	bool	Should the torrent be featured on the front page?
`free` *	int	Percentage (0-100) of the torrent's size that is free to leech
`fl_until` *	int	Number of days the torrent should offer freeleech
`doubleup` *	bool	Should the torrent offer double upload?
`du_until` *	int	Number of days the torrent should offer double upload
`sticky` *	bool	Should the torrent be stickied on the torrent index?
`mod_queue_opt_in` *	bool	Should the torrent be sent to moderation queue?

*Only available to staff and internal users.

Fetch a torrent

Endpoint: GET `/api/torrents/:id`

Example:

```
https://unit3d.site/api/torrents/39765?api_token=YOURTOKENHERE
```

Fetch torrents index (latest 25 torrents)

Endpoint: GET `/api/torrents`

Example:

```
https://unit3d.site/api/torrents?api_token=YOURTOKENHERE
```

Filter torrents

Endpoint: GET `/api/torrents/filter`

Optional Parameters:

Parameter	Type	Description
`perPage`	int	Amount of results to return per page (default: 25)
`sortField`	string	Field to sort by
`sortDirection`	string	Direction to sort the results. One of: `asc` (Ascending), `desc` (Descending) (default: `asc`)
`name`	string	Filter by the torrent's name
`description`	string	Filter by the torrent's description
`mediainfo`	string	Filter by the torrent's MediaInfo
`bdinfo`	string	Filter by the torrent's BDInfo

Parameter	Type	Description
`uploader`	string	Filter by the torrent uploader's username
`keywords`	string	Filter by any of the torrent's keywords (Multiple keywords can be comma-separated)
`startYear`	int	Return only torrents whose content was released after or in the given year
`endYear`	int	Return only torrents whose content was released before or in the given year
`categories`	int[]	Filter by the torrent's category
`types`	int[]	Filter by the torrent's type
`resolutions`	int[]	Filter by the torrent's resolution
`genres`	int[]	Filter by the torrent's genre
`tmdbId`	int	Filter by the torrent's TMDB ID
`imdbId`	int	Filter by the torrent's IMDB ID
`tvdbId`	int	Filter by the torrent's TVDB ID
`malId`	int	Filter by the torrent's MAL ID
`playlistId`	int	Return only torrents within the playlist of the given ID
`collectionId`	int	Return only torrents within the collection of the given ID
`free`	int	Filter by the torrent's freeleech discount (0-100)
`doubleup`	bool	Filter by if the torrent offers double upload
`featured`	bool	Filter by if the torrent is featured on the front page
`refundable`	bool	Filter by if the torrent is refundable
`highspeed`	bool	Filter by if the torrent has seeders whose IP address has been registered as a seedbox

Parameter	Type	Description
`internal`	bool	Filter by if the torrent is an internal release
`personalRelease`	bool	Filter by if the torrent's content is created by the uploader
`alive`	bool	Filter by if the torrent has 1 or more seeders
`dying`	bool	Filter by if the torrent has 1 seeder and has been downloaded more than 3 times
`dead`	bool	Filter by if the torrent has 0 seeders
`file_name`	string	Filter by the name of a file within a torrent
`seasonNumber`	int	Filter by the torrent's season number
`episodeNumber`	int	Filter by the torrent's episode number

Example:

```
https://unit3d.site/api/torrents/filter?
tmdbId=475557&categories[]=1&api_token=YOURTOKENHERE
```

Personal account info

Endpoint: GET `/api/user`

Response:

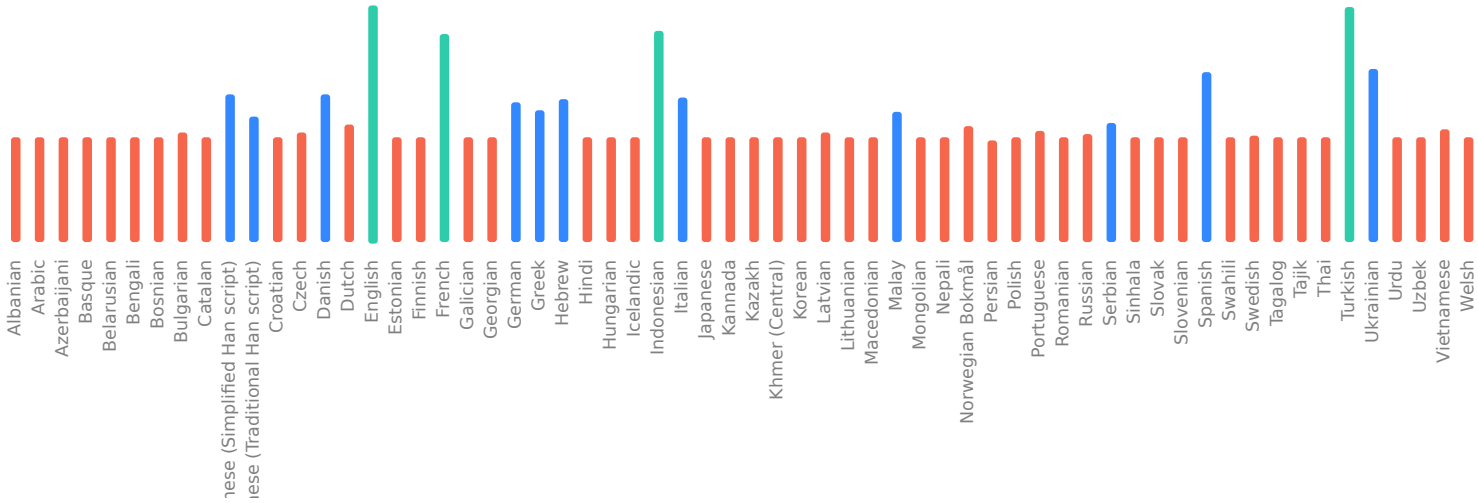

```
{"username":"UNIT3D","group":"Owner","uploaded":"50 GiB","downloaded":"1
GiB","ratio":"50","buffer":"124
GiB","seeding":0,"leeching":0,"seedbonus":"0.00","hit_and_runs":0}
```


Example:

```
https://unit3d.site/api/user?api_token=YOURTOKENHERE
```

Translations

UNIT3D uses weblate for translations. You can easily contribute to translations at https://hosted.weblate.org/engage/unit3d/. Use the following graphic to see if your native language could use some work.



Updating UNIT3D

Update UNIT3D to the latest version by reviewing the release notes and following the steps below:

1. Create backup

UNIT3D offers built-in backups. Refer to the Backups documentation for usage.

[!IMPORTANT] Ensure there is a complete backup before proceeding.

---

2. Enter maintenance mode

```
cd /var/www/html
php artisan down
```

3. Update UNIT3D

No se ha encontrado ningún vídeo que tenga un formato y tipo MIME compatibles.

0:00 / 0:00

[!NOTE] Before running the update, review the new release's minimum requirements to ensure the environment meets them.

1. **Proceed to update:**

   The updater will fetch the latest commits from the upstream repository and stage them for installation.

   ```
   cd /var/www/html
   php artisan git:update
   ```

   There will be a prompt to confirm each step; choose `yes` to overwrite with the new version.

```
Start the update process (yes/no) [yes]:
> yes
```

2. **Accept upstream files:**

   When prompted for each changed file, type `yes` to overwrite the local copy or press `Enter` to accept the default shown in brackets.

```
Update config/unit3d.php (yes/no) [yes]:
> yes

git checkout origin/master -- config/unit3d.php
[============================] (Done!)
```

3. **Run new migrations:**

```
Run new migrations (php artisan migrate) (yes/no) [yes]:
> yes
```

4. **Install new packages:**

```
Install new packages (composer install) (yes/no) [yes]:
> yes
```

5. **Compile assets:**

```
Compile assets (bun run build) (yes/no) [yes]:
> yes
```

Troubleshooting clean-up

The following commands are **optional** and should be run only as needed to resolve specific errors:

- **Finish any migrations not completed:**

```
sudo php artisan migrate
```

- **Reinstall dependencies:**

```
composer install --prefer-dist --no-dev -o
```

- **Clear caches:**

```
sudo php artisan cache:clear && \
sudo php artisan queue:clear && \
sudo php artisan auto:email-blacklist-update && \
sudo php artisan auto:cache_random_media && \
sudo php artisan set:all_cache
```

- **Rebuild static assets:**

```
sudo bun install && sudo bun run build
```

- **Restart services:**

```
sudo systemctl restart php8.4-fpm && \
sudo php artisan queue:restart && \
sudo php artisan up
```

- **If running external UNIT3D-Announce, restart the supervisor services:**

```
sudo supervisorctl reread && \
sudo supervisorctl update && \
sudo supervisorctl reload
```

Upgrading PHP version

Upgrade to PHP 8

```
sudo apt update  sudo apt -y upgrade
```

A reboot is important after any upgrade.

```
sudo systemctl reboot
```

After a few minutes SSH back into your server

```
sudo apt update  sudo apt install lsb-release ca-certificates apt-transport-https software-properties-common -y  sudo add-apt-repository ppa:ondrej/php
```

Hit enter key when prompted to add the repository

```
sudo apt update  sudo apt install php8.0  sudo apt-get install -qq curl debconf-utils php-pear php8.0-curl php8.0-dev php8.0-gd php8.0-mbstring php8.0-zip php8.0-mysql php8.0-xml php8.0-fpm php8.0-intl php8.0-bcmath php8.0-cli php8.0-opcache  sudo service apache2 stop
```

Next lets edit NGINX to use new PHP8

```
sudo nano /etc/nginx/sites-available/default
```

Find `fastcgi_pass unix:/var/run/php/***.sock;`

`***` will be your site name, unit3d or php7.4 for the most part

Replace `fastcgi_pass unix:/var/run/php/***.sock;` with `fastcgi_pass unix:/var/run/php/php8.0-fpm.sock;`.

Save and exit.

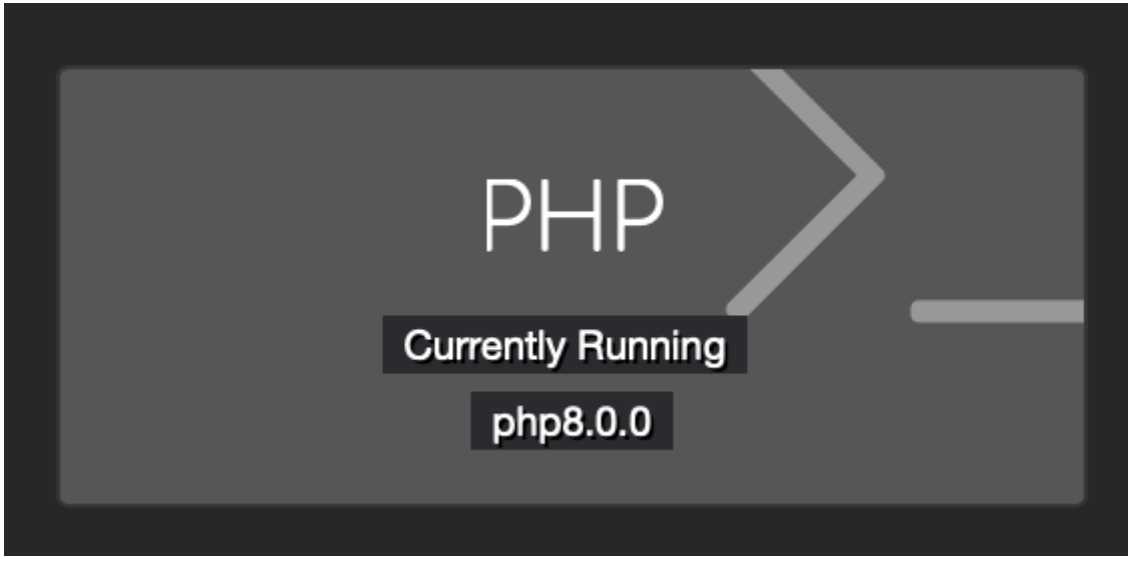
Test config `sudo nginx -t`

*If you didn't mess up you will see

```
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

```
sudo systemctl reload nginx  sudo systemctl reload php8.0-fpm  sudo apt purge '^php7.4.*'
```

You should now be running PHP8 and can confirm by checking your staff dashboard.



Upgrade to PHP 8.1

```
sudo apt update  sudo apt -y upgrade
```

A reboot is important after any upgrade.

```
sudo systemctl reboot
```

After a few minutes SSH back into your server

```
sudo apt update  sudo apt install lsb-release ca-certificates apt-transport-https software-properties-common -y  sudo add-apt-repository ppa:ondrej/php
```

Hit enter key when prompted to add the repository

```
sudo apt update  sudo apt install php8.1  sudo apt-get install -qq curl debconf-utils php-pear php8.1-curl php8.1-dev php8.1-gd php8.1-mbstring php8.1-zip php8.1-mysql php8.1-xml php8.1-fpm php8.1-intl php8.1-bcmath php8.1-cli php8.1-opcache  sudo service apache2 stop
```

Next lets edit NGINX to use new PHP 8.1

```
sudo nano /etc/nginx/sites-available/default
```

Find `fastcgi_pass unix:/var/run/php/***.sock;`

`***` will be your site name, unit3d or php8.0 for the most part

Replace `fastcgi_pass unix:/var/run/php/***.sock;` with `fastcgi_pass unix:/var/run/php/php8.1-fpm.sock;` .

Save and exit.

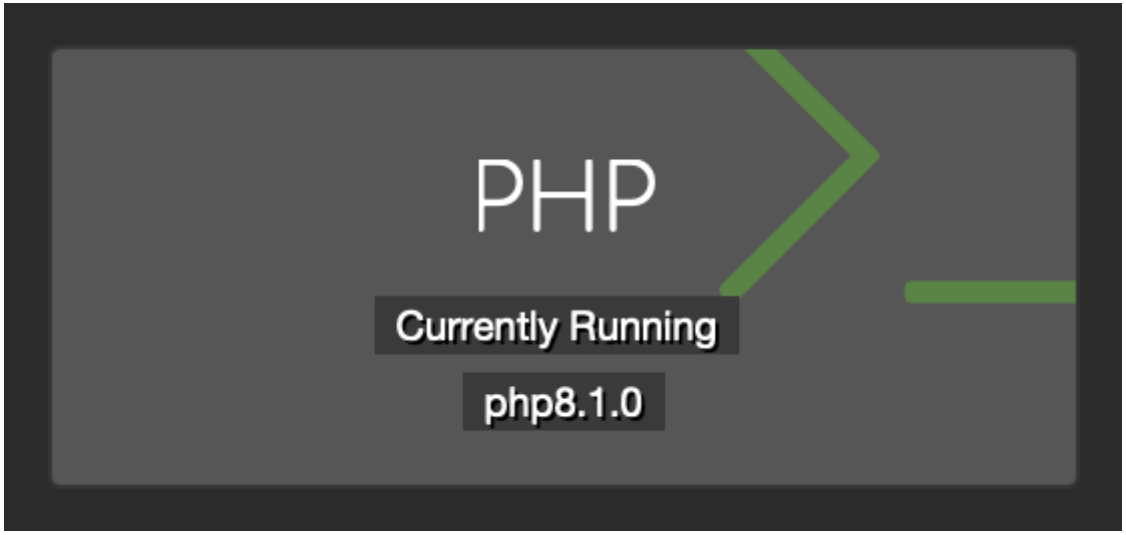
Test config `sudo nginx -t`

*If you didn't mess up you will see

```
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

```
sudo systemctl reload nginx  sudo systemctl reload php8.1-fpm  sudo apt purge '^php7.4.*'
```

You should now be running PHP8.1 and can confirm by checking your staff dashboard.



Upgrade to PHP 8.2

```
sudo apt update  sudo apt -y upgrade
```

A reboot is important after any upgrade.

```
sudo systemctl reboot
```

After a few minutes SSH back into your server

```
sudo apt update  sudo apt install lsb-release ca-certificates apt-transport-https software-properties-common -y  sudo add-apt-repository ppa:ondrej/php
```

Hit enter key when prompted to add the repository

```
sudo apt update  sudo apt install php8.2  sudo apt-get install -qq curl debconf-utils php-pear php8.2-curl php8.2-dev php8.2-gd php8.2-mbstring php8.2-zip php8.2-mysql php8.2-xml php8.2-fpm php8.2-intl php8.2-bcmath php8.2-cli php8.2-opcache  sudo service apache2 stop  sudo apt remove apache2
```

Next lets edit NGINX to use new PHP 8.2

```
sudo nano /etc/nginx/sites-available/default
```

Find `fastcgi_pass unix:/var/run/php/***.sock;`

`***` will be your site name, unit3d or php8.1 for the most part

Replace `fastcgi_pass unix:/var/run/php/***.sock;` with `fastcgi_pass unix:/var/run/php/php8.2-fpm.sock;`.

Save and exit.

Test config `sudo nginx -t`

*If you didn't mess up you will see

```
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

`sudo systemctl restart nginx` `sudo systemctl restart php8.2-fpm` `sudo systemctl stop php8.1-fpm` `sudo apt purge '^php8.1.*'` `sudo apt autoremove`

You should now be running PHP8.2 and can confirm by checking your staff dashboard.

NOTE: If you had tuning done on PHP 8.1 you will need to reapply them to new PHP 8.2 configs.
`sudo nano /etc/php/8.2/fpm/pool.d/www.conf` `sudo nano /etc/php/8.2/fpm/php.ini`

Upgrade to PHP 8.3

Save existing php package list to packages.txt file in case you have some additional ones not noted in this guide.

`sudo dpkg -l | grep php | tee packages.txt`

Add Ondrej's PPA

`sudo add-apt-repository ppa:ondrej/php` # Press enter when prompted. `sudo apt update`

Install new PHP 8.3 packages

`sudo apt install php8.3-common php8.3-cli php8.3-fpm php8.3-{redis,bcmath,curl,dev,gd,igbinary,intl,mbstring,mysql,opcache,readline,xml,zip}`

Next lets edit NGINX to use new PHP 8.3

`sudo nano /etc/nginx/sites-available/default`

Find `fastcgi_pass unix:/var/run/php/***.sock;`

`***` will be your site name, unit3d or php8.2 for the most part

Replace `fastcgi_pass unix:/var/run/php/***.sock;` with `fastcgi_pass unix:/var/run/php/php8.3-fpm.sock;`.

Save and exit.

Test config `sudo nginx -t`

*If you didn't mess up you will see

```
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

`sudo systemctl restart nginx` `sudo systemctl restart php8.3-fpm` `sudo systemctl stop php8.2-fpm`

Remove old packages

`sudo apt purge '^php8.2.*'`

You should now be running PHP8.3 and can confirm by checking your staff dashboard.

NOTE: If you had tuning done on PHP 8.2 you will need to reapply them to new PHP 8.3 configs.
`sudo nano /etc/php/8.3/fpm/pool.d/www.conf` `sudo nano /etc/php/8.3/fpm/php.ini`

UNIT3D-Announce

UNIT3D-Announce is an optional external announce service written in rust that is designed to reduce CPU usage on high-volume trackers managing over 1 million peers.

[!TIP] Enable only when extra performance is needed. The default announce suffices otherwise.

Refer to the upstream documentation for configuration and usage details.

- **Consult** the README for installation and usage instructions.
- **Review** the `.env.example` to configure required variables.